Bangalore Institute of Technology

An Autonomous Institution Under VTU
K.R Road, V.V Puram, Bangalore – 560004

Board of Computer Science (CS&E /IS&E /AI&ML /CSE-DS /CSE-ICB)

AI MODULEWISE QUESTION BANK -2026-27 (ODD)
Batch: 2024

Course Title : Artificial Intelligence | **Course Code :BCS15B**
No. of Lecture Hrs./Week : 3:0:0:0 | **Exam Hours : 03**
CIE Marks:50 | **SEE Marks : 50**

Module -1 Questions.

1. Define Artificial Intelligence. Discuss the four aspects for AI.
2. Discuss the state-of-the-art applications of Artificial Intelligence.
3. Is AI a science, or is it engineering or neither or both? Explain.
4. Examine the AI literature to discover whether the following tasks can currently be solved by computers.
   i) Playing a decent game of table tennis (ping-pong)
   ii) Discovering and proving new mathematical theorems
   iii) Giving competent legal advice in a specialized area of law
   iv) Performing a complex a surgical operations
5. What are agents? Define the terms percepts, actions, goals and environment with reference to the following:
   a) Automated taxi driving
   b) Medical diagnosis system
   c) Satellite image analysis
   d) Part picking robot
6. What are the characteristics of rational agents?
7. Describe PEAS with respect to following agent types:
   a) Automated taxi driving
   b) Medical diagnosis system
   c) Satellite image analysis
   d) Part picking robot
   e) Playing a tennis match
   f) Performing a high jump
   g) Bidding on an item in an auction.
   h) Playing soccer.
   i) Exploring the subsurface oceans of Titan.
   j) Shopping for used AI books on the Internet.
   k) Knitting a sweater.

l) Bidding on an item at an auction

8. Define the following :
   i) Artificial Intelligence ii) Agent iii)rational agents iv) Rationality v) Logical reasoning.

9. Summarize the properties of task environment? Give examples

10. What are agent programs? Outline structure of different agent programs listed below and its properties with suitable block diagrams for each.

    a. Table-driven agent
    b. Simple reflex agent
    c. Reflex agent with internal state
    d. Agent with explicit goals
    e. Utility-based agent

11. Implement a simple reflex agent for the vacuum environment. Run the environment with this agent for all possible initial dirt configurations and agent locations. Record the performance score for each configuration and the overall score.

12. Identify the characteristics satisfied by each agent and Justify your answer.

	Accessible	Deterministic	Episodic	Static	Discrete
Solitaire					
Part-Picking Robot					
Taxi driving					
Online shopping					
Medical diagnosis					

13. Write pseudocode agent programs for the goal based and utility based agents.

14. With neat diagram explain general learning agent.

15. Mention the different ways to represent states and transitions between them.

Solutions for fewLast Year Question Paper Questions is in the Following Link:

https://share.google/NA72v6SuZmyuvAa46

Faculty In-charges

Course Coordinator **Module Coordinator** **IQAC** **Program Coordinator**